Exp 12 · MOSFET inverter (resistive load)

Manual pp. 34–35 · LTspice · CO4 · theory: ECE2101 unit 4 (MOSFETs) · links straight to the Digital Electronics lab (logic gates)

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What you already have:

- From Exp 9, a FET's gate voltage controls its channel.
- From Exp 5, a transistor can be a switch: cut-off = open, saturation = closed.
- From digital electronics: NOT gate, logic 1 and logic 0.

Three terms first:

- **Enhancement nMOS:** a MOSFET whose gate is insulated by oxide (so it draws **no** gate current). It has **no** channel at $V_{GS} = 0$. A channel forms only when $V_{GS} > \mathbf{V_{TH}}$ (threshold voltage).
- **Inverter:** a NOT gate. Input high gives output low, and input low gives output high.
- **VTC (voltage transfer characteristic):** V_{out} plotted against V_{in}.

MANUAL EXP 12, AIM

Analyse an inverter circuit using a MOSFET. Viva form: "Explain the two states of the circuit, and mark V_{OH}, V_{OL}, V_{IL}, V_{IH} and V_M on the VTC."

GUESS FIRST

When the input is 0 V, is the MOSFET on or off? What is the output then?

How it works

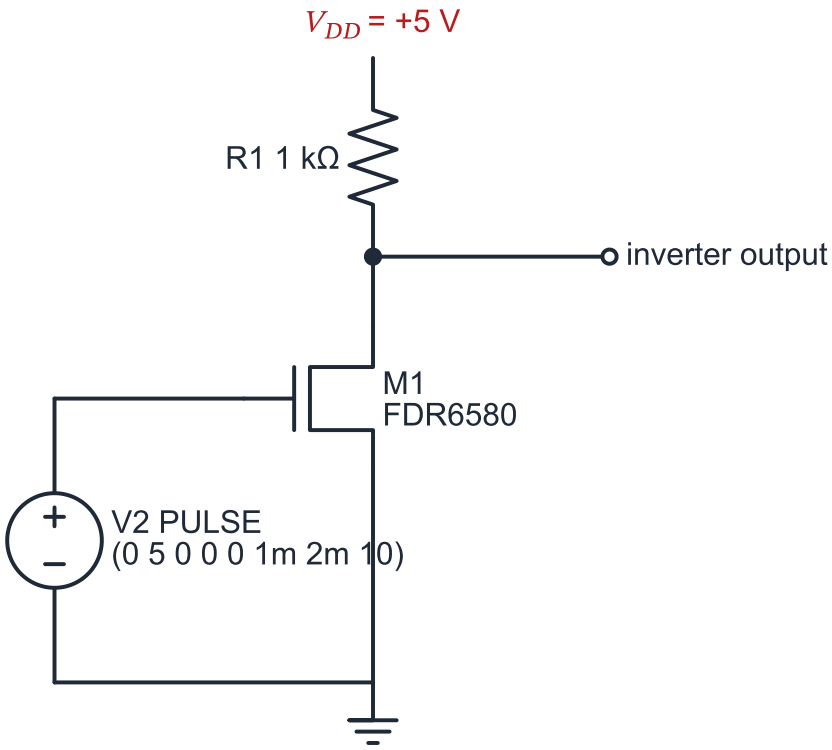


Fig 12.1, redrawn: 1 kΩ from +5 V to the drain (output); input pulse on the gate; source to ground.

Input	MOSFET	Current	Output
0 V (logic 0), below V_{TH}	**off** (cut-off): no channel	I_D = 0, so no drop across the 1 kΩ	$V_{out} = V_{DD}$ = **5 V (logic 1)**
5 V (logic 1), above V_{TH}	**on**, fully (deep triode): the channel is a very low resistance (≈ 9 mΩ for this part)	I_D ≈ 5 V/1 kΩ = 5 mA	V_{out} ≈ 5 mA × 9 mΩ ≈ **0 V (logic 0)**

VTC terms:

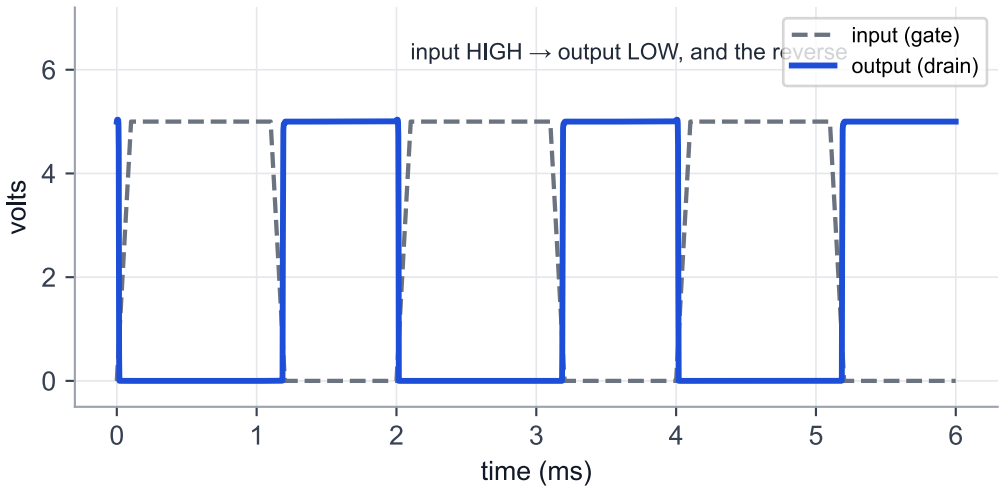
Term	Meaning
V_{OH} / V_{OL}	output high / output low levels
V_{IL}	the highest input still read as 0. Where the VTC slope first reaches −1.
V_{IH}	the lowest input read as 1. Where the slope returns to −1.
V_M	switching threshold, where $V_{out} = V_{in}$
noise margins	$NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$: how much noise an input can carry before the gate misreads it

Do it

DO IT (LTSPICE)

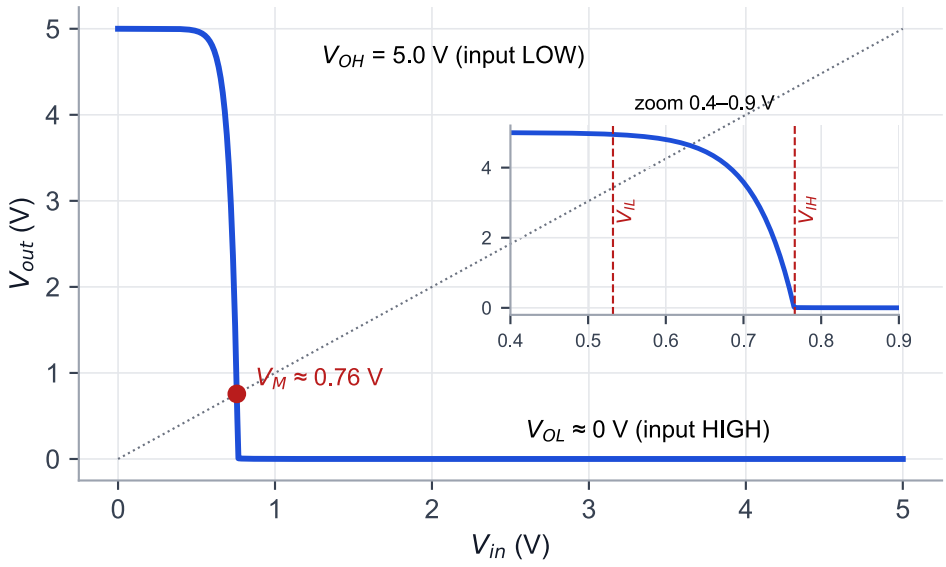
1. Draw Fig 12.1: M1: Pick New MOSFET → **FDR6580**; R1 = 1k; V1 = 5 V.
2. V2: **PULSE**: V1 = 0, V2 = 5, Tdelay = 0, Trise = 0, Tfall = 0, Ton = 1m, Tperiod = 2m, Ncycles = 10, which gives `PULSE(0 5 0 0 0 1m 2m 10)`.
3. Directive: `.tran 0 20m 0 1u`. The manual's bare `.tran 20m` works too, but its coarse steps show slanted edges and a false spike up to about 6.8 V.
4. Plot the input and output. Set the y-axis limits manually to −0.5 … 5.5 V. Read V_{min} and V_{max} of each.
5. **VTC (extra marks):** replace the pulse with a DC source, run `.dc V2 0 5 0.001`, and plot V(out) against V2.

What you should see



Transient: the output is the input upside-down: 5 V when the input is 0, about 0 V when the input is 5 V.

Voltage	V_{min}	V_{max}
V_{in}	0 V	5 V
V_{out}	≈ 0 V (0.04 mV)	5.0 V



VTC from LTspice: a very sharp switch near 0.76 V (zoom inset).

V_{OH}	V_{OL}	V_{IL}	V_{IH}	V_M	NM_L	NM_H
5.00 V	≈ 0 V	0.53 V	0.77 V	0.76 V	0.53 V	4.23 V

Why it switches so low and so sharply: the FDR6580 is a **power** MOSFET. It conducts amps from a tiny gate overdrive, so just past V_{TH} it already pulls the 5 mA that the 1 kΩ allows. The switch therefore happens right at V_{TH}, far below $V_{DD}/2$.

The manual's Fig 12.2 (with $V_{IL} \approx 0.9$ V and $V_{IH} \approx 2.2$ V) is a generic textbook VTC, not this part's.

TRAP: V_{TH} VS V_M

The manual says "V_{th} is the inverter threshold voltage, which is $V_{dd}/2$". These are two different quantities:

- **V_{TH}** is the MOSFET's own turn-on voltage.
- **V_M** is the inverter's switching point (where $V_{out} = V_{in}$).

$V_M = V_{DD}/2$ is the **design goal** of a symmetric CMOS inverter. In this resistor-load circuit, $V_M \approx 0.76$ V.

WRITE IN YOUR RECORD

Result: the output is inverted with respect to the input, with maximum voltage **5 V** as logic 1 and minimum voltage **≈ 0 V** as logic 0. (VTC: $V_{IL} = 0.53$ V, $V_{IH} = 0.77$ V, $V_M = 0.76$ V.)

Theory for the record: the inverter is the basic digital gate; NAND, NOR and larger logic are built from the same idea. In the resistive-load nMOS inverter, an enhancement nMOS pulls the output down and a resistor pulls it up to V_{DD}. For $V_{in} < V_{TH}$ the MOSFET is cut off, no current flows, and $V_{out} = V_{DD}$ (logic 1). For a high input the MOSFET conducts in the linear region and pulls the output close to 0 (logic 0). The VTC shows V_{OH}, V_{OL} and the transition region, with V_{IL} and V_{IH} at the unity-slope points and V_M where $V_{out} = V_{in}$. Noise margins are $NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$. A drawback: current (and power) flows continuously when the output is low, which CMOS removes.

CHECK 1

(a) How much power does this inverter waste when its output is LOW? When HIGH? (b) Why does the MOSFET's gate draw no DC current? (c) What replaces the resistor in a CMOS inverter, and what does that fix?

Answers

Opening question. Input 0 V: the MOSFET is **off**, so no current flows and the output = **5 V**. Input 5 V: the MOSFET is **on** and the output ≈ 0 V. V_{OH} = 5 V, V_{OL} ≈ 0, V_{IL} = 0.53 V, V_{IH} = 0.77 V, V_M = 0.76 V.

Check 1. (a) LOW: 5 mA flows from the 5 V supply, so P = 5 V × 5 mA = **25 mW**, nearly all in the resistor. HIGH: no current, so ≈ 0 W. (b) The gate is insulated from the channel by a thin oxide layer: a capacitor, which passes no DC. (c) A **pMOS** transistor (pull-up). Only one transistor is on in each state, so no DC current flows from V_{DD} to ground. The static power becomes ≈ 0, and V_{OL} is exactly 0.